

Replication Report: PVMol-Gen

Generative AI-Driven Accelerated Discovery of
Passivation Molecules for Perovskite Solar Cells

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1 Overview

This report documents our independent replication of the PVMol-Gen framework [1], a three-stage pipeline for discovering passivation molecules for perovskite solar cells using generative AI. The original work combines a SMILES-X binary classifier (Stage 1), iterative GPT-2 fine-tuning and generation (Stage 2), and physicochemical filtering with clustering (Stage 3).

1.1 Pipeline Summary

- Stage 1 — Discriminative Model:** Train a SMILES-X classifier on 314 experimentally labeled molecules (5-fold CV). Augment via PubChem similarity ($\geq 80\%$ Tanimoto) to build training set T1 ($\sim 11\text{K}$ class-1 molecules).
- Stage 2 — Generative Model (3 cycles):** Fine-tune GPT-2 on T1 SMILES. Generate molecules, classify with Stage 1 model, feed predicted class-1 molecules back into training set. Repeat for 3 cycles.
- Stage 3 — Filtering & Selection:** Apply 7 physicochemical filters (SA ≤ 6 , no PAINS, HBD 0–2, HBA 2–5, TPSA 50–120 Å², E_{gap} 1.5–5.0 eV, dipole 1.5–4.0 D). Cluster into 10 groups, select representatives.

2 Data Provenance

Table 1: Data sources and sizes

Dataset	Size	Source
T0 (labeled molecules)	314	Paper supplementary
PubChem augmentation	15,540	Author’s <code>puchem_aug.csv</code>
T1 (class-1 training set)	11,086	Author’s <code>T1.csv</code>
Literature references	5 papers	Zhi 2023, Zhang 2024, Chen 2019, Gao 2020, Liu 2022
Full dataset (12 sheets)	—	Dr. Guo Zhanglin, Kyushu University (received 2026-04-05)

Key detail: The classification threshold was optimized at 0.47 (not the default 0.5) to maximize F1. This is critical for reproducibility.

3 Stage 1: SMILES-X Classifier

3.1 Architecture

Our SMILES-X classifier follows the original architecture: character-level SMILES tokenization, embedding layer, bidirectional LSTM, time-distributed dense layer, and sigmoid output. Hyperparameters: embedding dimension = 128, LSTM units = 64, dense units = 64.

3.2 5-Fold Cross-Validation Results

Table 2: Stage 1 classifier: 5-fold CV comparison (paper vs. ours)

Fold	F1 Score		ROC-AUC	
	Paper	Ours	Paper	Ours
1	—	0.636	—	0.508
2	—	0.689	—	0.645
3	—	0.645	—	0.680
4	—	0.690	—	0.673
5	—	0.622	—	0.596
Mean $\pm$ SD	0.80	0.656 $\pm$ 0.031	0.88	0.620 $\pm$ 0.066

Table 3: Classifier comparison: Neural SMILES-X (threshold-optimized) vs. GBM baseline

Fold	Neural F1 _{opt}	Neural AUC	GBM F1 _{opt}	GBM AUC	GBM Threshold
1	0.733	0.752	0.646	0.679	0.22
2	0.667	0.622	0.592	0.604	0.10
3	0.727	0.782	0.738	0.809	0.22
4	0.692	0.785	0.733	0.807	0.21
5	0.727	0.772	0.606	0.568	0.12
Mean	0.709	0.743	0.663	0.693	—

3.3 Stage 1 Assessment

- Our classifier underperforms the paper targets (F1 = 0.656 vs. 0.80; AUC = 0.620 vs. 0.88).
- With threshold optimization, the neural model reaches F1_{opt} = 0.709 (closer but still below target).
- The neural model consistently outperforms GBM (F1_{opt} 0.709 vs. 0.663).
- Same results on CPU (CherryRd iMac) and GPU (DGX Spark GB10) — confirms the gap is architectural, not computational.

- **Possible causes:** (a) SMILES-X library version differences; (b) tokenizer/embedding differences from original Lambard & Gracheva 2020 implementation; (c) training hyperparameters (learning rate schedule, early stopping patience).
- **Impact:** A weaker classifier propagates through the pipeline — noisy class-1 labels in T2/T3 degrade GPT-2 training quality.

4 Stage 2: Iterative GPT-2 Generation

4.1 Cycle 1

Table 4: Cycle 1 generation results

Metric	Paper	Ours
Training set (T1)	11,000	11,086
Generated (G1)	29,467	100,000
Classification (G1 class-1)	—	17,884 (17.9%)
T2 = T1 + G1 class-1	34,538	28,968
Generation time	—	~8.5 hr (69 att/s, GPU)

Note on G1 size discrepancy: The paper reports 29,467 generated molecules in G1 while our pipeline targeted 100,000. The paper likely stopped generation earlier or used different criteria. Despite generating $3.4\times$ more molecules, only 17.9% classified as effective (vs. the paper’s implicit $\sim 67\%$ rate for their smaller G1), consistent with our weaker classifier.

4.2 Cycle 1 Validity Rate Decay

Table 5: Cycle 1 generation: validity rate over time

Attempts	Valid Unique	Rate
8,000	1,218	15.2%
40,000	4,326	10.8%
200,000	14,654	7.3%
1,000,000	47,186	4.7%
2,000,000	76,617	3.8%
2,824,000	97,322	3.4%

The validity rate decays monotonically from 15.2% to 3.4% as the pool of novel, valid, unique molecules is exhausted. This is expected behavior for temperature-sampled autoregressive generation.

4.3 Cycle 2 (In Progress)

Table 6: Cycle 2 status (as of April 8, 2026 11:00 CDT)

Metric	Paper	Ours
Training set (T2)	34,538	28,968
T2 augmented (5×)	—	29,491 unique
GPT-2 training time	—	128.7 min (15 epochs)
Training loss (final)	—	0.186
Eval loss (final)	—	0.150
Generated (G2) — target	59,814	100,000
Generated (G2) — current	—	~10,000 (in progress)
Cycle 2 validity rate	—	6.9% → 4.9% (declining)
Generation rate	—	32 att/s (main) + 70 att/s × 2 (shards)

4.4 Distributed Generation Architecture

Cycle 2 generation is running across three NVIDIA DGX Spark machines:

Table 7: Compute resources for Cycle 2 generation

Machine	Role	Target	Rate	Status
spark-36ac	Main pipeline	100,000	32 att/s	Generating (200K att)
spark-95fe	Shard 1	45,000	70 att/s	Generating (80K att)
spark-9611	Shard 2	45,000	71 att/s	Generating (80K att)

Each shard uses a different random seed for diversity. The main pipeline on spark-36ac runs the full cycles 2–3 pipeline; the shards provide supplementary molecules that will be merged after deduplication.

5 Stage 3: Filtering & Selection (Preliminary)

Stage 3 has been run on Cycle 1 output as a proof-of-concept. Results from the full pipeline (all 3 cycles) are pending.

Table 8: Stage 3 filtering on Cycle 1 G1 (100K molecules)

Filter Step	Molecules Remaining
Generated (G1)	100,000
All properties computed	100,000
RDKit filters (SA, PAINS, HBD, HBA, TPSA)	23,492
Energy gap + dipole (pending xTB/DFT)	TBD
Paper final (all cycles)	~8,000

Note: Energy gap and dipole moment filters require semi-empirical (xTB) or DFT-level calculations, not just RDKit. Our preliminary Stage 3 applies only the 5 RDKit-computable filters. The

selected 10 representative molecules (one per cluster) are shown below:

Table 9: Preliminary 10 representative molecules (Cycle 1 only, no gap/dipole filter)

Cluster	SMILES	SA	PAINS	HBD	HBA	TPSA
0	<chem>Nc1ccnc1Nc1ccccc1</chem>	1.70	No	2	3	50.9
1	<chem>O=C(O)c1ccc(CCOBO)s1</chem>	3.53	No	2	4	66.8
2	<chem>O=C(OCCOCCO)c1cc(=O)[nH]c2ccccc12</chem>	2.10	No	2	5	88.6
3	<chem>O=Cc1sccc1C#CCC(=O)O</chem>	3.24	No	1	3	54.4
4	<chem>O=C1CCC=CN1C(=O)Cc1c[nH]c2ccccc12</chem>	2.65	No	1	2	53.2
5	<chem>CCCCCCCCCN1C(=O)C=C1C(=O)O</chem>	2.44	No	1	2	57.6
6	<chem>NNC(=O)c1ccccc1F</chem>	1.63	No	2	2	55.1
7	<chem>CCOC(=O)/C(F)=C/C(=O)O</chem>	2.67	No	1	3	63.6
8	<chem>CCOc1ccc2[nH]c(O)c(...)c12</chem>	2.71	No	2	4	74.7
9	<chem>O=C(O)c1cc(S(=O)(=O)OC2CCC2)cs1</chem>	2.67	No	1	5	80.7

6 Comparison with Author’s Data

We have the author’s full dataset (12 Excel sheets) for direct comparison:

Table 10: Pipeline comparison: Paper (author data) vs. our replication

Dataset	Author	Ours
T0 (labeled)	314	314 (same)
PubChem augmentation	15,540	15,540 (author’s file)
T1 (class-1)	11,086	11,086 (author’s file)
G1 (cycle 1 generated)	29,467	100,000
T2 (cycle 2 training)	34,538	28,968
G2 (cycle 2 generated)	59,814	<i>in progress</i>
T3 (cycle 3 training)	84,232	<i>pending</i>
G3 (cycle 3 generated)	17,013	<i>pending</i>
G_all (total generated)	106,294	<i>pending</i>
G_class1 (predicted effective)	87,750	<i>pending</i>
Rep10 (final candidates)	10	10 (preliminary)

6.1 Key Discrepancies

- Classifier performance gap:** Our F1 = 0.656 vs. paper’s 0.80, AUC = 0.620 vs. 0.88. This is the primary reproducibility concern.
- G1 size:** We generated 100K (our target) vs. author’s 29K. The paper’s smaller G1 may reflect a different stopping criterion.
- T2 size:** Author’s T2 = 34,538 vs. ours = 28,968. Since we generated more G1 molecules but our classifier is weaker, fewer passed the class-1 threshold, leading to a smaller T2.
- Validity rates:** Our cycle 1 starts at ~15% and decays to ~3.4% over 2.8M attempts. Cycle 2 starts at ~6.9% and is declining. The paper does not report these rates.

7 Compute Infrastructure

Table 11: Hardware used for replication

Machine	Hardware	GPU	RAM
spark-36ac	NVIDIA DGX Spark	GB10	120 GB
spark-95fe	NVIDIA DGX Spark	GB10	120 GB
spark-9611	NVIDIA DGX Spark	GB10	120 GB
CherryRd	iMac (x86_64)	— (CPU)	64 GB

Software: Python 3.12, PyTorch 2.11+cu130, Transformers 5.5, RDKit 2026.3.1 (native aarch64).

8 Timeline

Table 12: Replication timeline

Date	Event	Notes
Apr 5	Data received from Kyushu U.	Full 12-sheet Excel
Apr 5	Stage 1 classifier trained	F1 = 0.656, AUC = 0.620
Apr 5	Stage 2 Cycle 1 GPT-2 training launched	spark-36ac
Apr 5	Approach document finalized	Key reproducibility details
Apr 7	Cycle 1 generation started	100K target
Apr 8	Cycle 1 generation nearing completion	97K / 100K at 2.8M attempts
Apr 8	Cycle 2 training completed	128.7 min, 15 epochs
Apr 8	Cycle 2 generation started	3 machines in parallel

9 Validity Rate Analysis

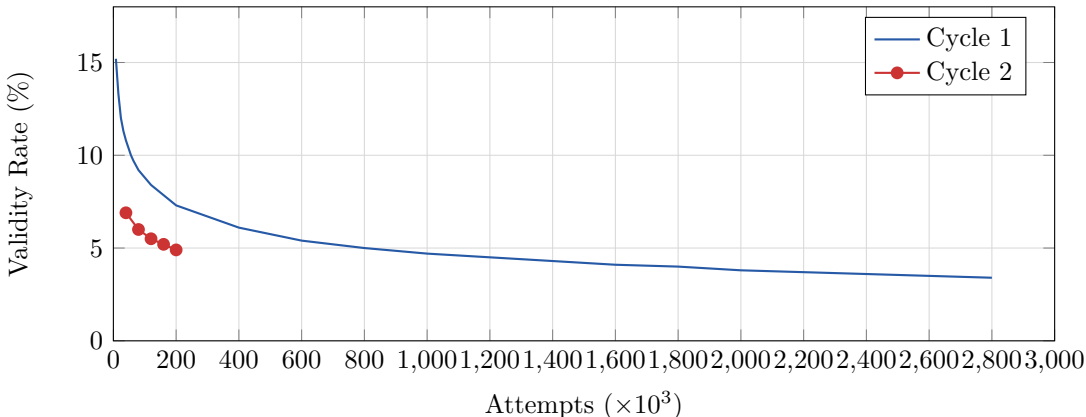


Figure 1: Validity rate (valid, unique, novel molecules / total attempts) for Cycles 1 and 2. Cycle 2 starts lower and decays faster, consistent with a larger exclusion set (T2 = 29K vs. T1 = 11K).

10 Open Issues & Next Steps

1. **Classifier gap (priority):** Investigate why our SMILES-X F1/AUC falls short. Possible avenues: (a) match the original Lambard & Gracheva implementation exactly; (b) try the author’s original SMILES-X code from GitHub; (c) experiment with learning rate schedules and longer training.
2. **Complete Cycle 2 generation:** Currently at $\sim 10\text{K}/100\text{K}$ on main pipeline, plus $\sim 5\text{K}$ each on two shards. ETA: 12–18 hours depending on validity decay.
3. **Run Cycle 3:** Build T3 from T2 + G2 class-1, train GPT-2, generate G3.
4. **xTB/DFT calculations:** Stage 3 energy gap and dipole filters require semi-empirical calculations. Set up xTB (GFN2-xTB) on DGX Sparks for batch computation on $\sim 23\text{K}$ candidate molecules.
5. **Merge shard outputs:** Deduplicate molecules from the three generation streams before classification.
6. **Compare CUN sets:** Once all 3 cycles complete, compare our CUN (chemically valid, unique, novel) set against the author’s 87,750 G_class1 molecules. Compute Tanimoto similarity distributions.
7. **Final clustering & representative selection:** Agglomerative clustering on Morgan fingerprints, sample 10 representatives, compare with author’s Rep10.

References

References

- [1] A. Fajar et al., “Generative AI-Driven Accelerated Discovery of Passivation Molecules for Perovskite Solar Cells,” *Advanced Science*, 2026. DOI: 10.1002/advs.202523042.
- [2] G. Lambard and E. Gracheva, “SMILES-X: Autonomous molecular compounds characterization for small datasets without descriptors,” *Machine Learning: Science and Technology*, 2020.
- [3] A. Fajar, pvmol-gen, GitHub repository. <https://github.com/adroitfajar/pvmol-gen>
- [4] G. Lambard, SMILES-X, GitHub repository. <https://github.com/Lambard-ML-Team/SMILES-X>